Day 095 | Unit 07 | Teacher plan

Understand and Use Percent

Connect Percent Models and Benchmarks

Standards: 6.RP.A.3c

Learning target: I can connect Percent Models and Benchmarks, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 095 | Unit 07 | Lesson

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01 / Learn

Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

Write 10% as a simplified fraction and decimal.

1. Percent means per 100.

2. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal.

3. 10% = 10/100 = 1/10

4. divide by 100 for 0.1.

5. State the result in the form requested: 1/10; 0.1.

Answer: 1/10; 0.1

Worked example 2

A school club has 60 members; 30% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 30% as 3/10.

3. Multiply by 60 to obtain 18.

4. Compare the rate with 1.

Answer: 18 participants; n = (30/100) × 60; less than the whole because the percent is below 100

Day 095 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 50% as a simplified fraction and decimal.

02 / Constructed response

Find 10% of 240.

03 / Constructed response

Write 20% as a simplified fraction and decimal.

04 / Constructed response

Find 125% of 140.

05 / Constructed response

Write 25% as a simplified fraction and decimal.

06 / Constructed response

Find 20% of 200.

07 / Constructed response

Write 125% as a simplified fraction and decimal. Explain why the decimal can be greater than 1.

08 / Numeric entry and equation

A school club has 40 members; 15% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

09 / Multiple select

Select all representations of 375%.
A. 375
B. 375/100
C. 375/10
D. 3.75

10 / Constructed response

Find 15% of 120. Then explain a mental-math or fraction strategy that checks your answer.

Day 095 | Unit 07 | Answer key

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1. 1/2; 0.5

50% = 50/100 = 1/2; divide by 100 for 0.5.

6.RP.A.3c | Constructed response

2. 24

10/100 × 240 = 24.

6.RP.A.3c | Constructed response

3. 1/5; 0.2

20% = 20/100 = 1/5; divide by 100 for 0.2.

6.RP.A.3c | Constructed response

4. 175

125/100 × 140 = 175.

6.RP.A.3c | Constructed response

5. 1/4; 0.25

25% = 25/100 = 1/4; divide by 100 for 0.25.

6.RP.A.3c | Constructed response

6. 40

20/100 × 200 = 40.

6.RP.A.3c | Constructed response

7. 5/4; 1.25

125% = 125/100 = 5/4; divide by 100 for 1.25.

6.RP.A.3c | Constructed response

8. 6 participants; n = (15/100) × 40; less than the whole because the percent is below 100

Rename 15% as 3/20. Multiply by 40 to obtain 6. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

9. B, D - 375/100; 3.75

Percent means per 100. Divide 375 by 100 to get 3.75; simplify 375/100 to 15/4.

6.RP.A.3c | Multiple select

10. 18

15/100 × 120 = 18.

6.RP.A.3c | Constructed response